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01211373 Machine Learning and Programming for Industry

Week 1 — Foundations & Tooling

3rd-Year Undergraduate • scikit-learn & PyTorch • Fall Semester

Why Machine Learning Belongs in This Program

Three problems classical control and rule-based automation struggle with on their own:

Predictive Maintenance

Spotting the early signature of bearing wear or motor faults in vibration and current data, before failure.

Visual Inspection

Flagging defective parts on a line from images, faster and more consistently than manual QA.

Learning-Based Control

Handling contact-rich or hard-to-model dynamics that resist a clean analytical controller.

Where We're Headed: 14 Weeks

WEEKS 1–6

Classical Machine Learning

scikit-learn • regression, classification, ensembles, PCA, clustering, evaluation

WEEKS 7–14

Deep Learning

PyTorch • neural nets, CNNs, LSTMs, attention, reinforcement learning, deployment

Wk 6

Project Checkpoint 1

Wk 9

First CNN Lab

Wk 12

RL on a Simulated Robot

Wk 14

Capstone Presentations

Where Does ML Fit Next to Classical Control?

MODEL-BASED

Classical Control & Estimation

- Built from known physics (dynamics, kinematics)
- PID, LQR, Kalman/particle filters
- Strong guarantees when the model is accurate
- Struggles with unmodeled or highly nonlinear effects

DATA-DRIVEN

Machine Learning

- Learns patterns directly from example data
- Classification, regression, clustering, RL
- Handles nonlinearities that are hard to derive by hand
- Needs good data, and its guarantees are statistical

This course treats ML as a complement to control theory, not a replacement for it.

What Is Machine Learning?

A system that improves its performance on a task by learning patterns from data, rather than by following rules a human wrote down explicitly.

Data

sensor logs, images, labels

Algorithm

e.g. gradient boosting, a neural net

Model

learned function f(x)

Predictions

fault / no-fault, torque estimate...

This is the roadmap every lab this semester will follow, whatever the algorithm.

Three Flavors of Machine Learning

Supervised Learning

Learn from labeled examples — every training input comes with the correct output.

Unsupervised Learning

Find structure in data that has no labels — groupings, compressed representations.

Reinforcement Learning

Learn by interacting with an environment, guided by reward rather than a fixed answer key.

Supervised Learning: Two Flavors

DISCRETE OUTPUT

Classification

"Is this bearing reading Normal or Faulty?"

Output: one of a fixed set of classes

CONTINUOUS OUTPUT

Regression

"What is the estimated joint torque from these strain-gauge readings?"

Output: a real-valued number

Unsupervised Learning: Finding Structure

Clustering

Group similar vibration signatures together, with no fault labels at all.

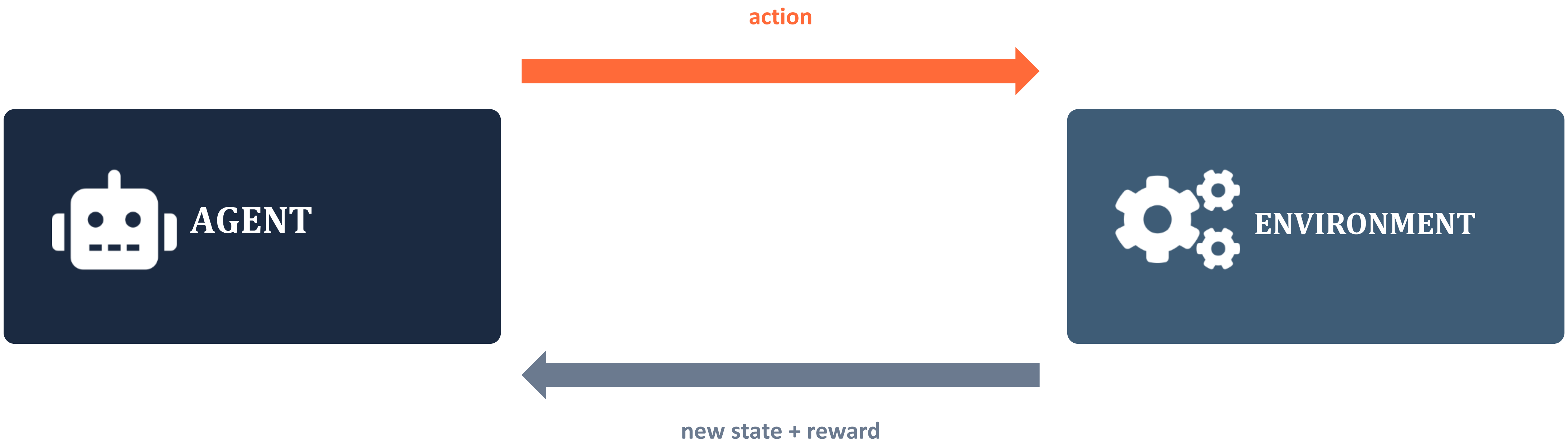
e.g. k-means, DBSCAN

Dimensionality Reduction

Compress dozens of correlated sensor channels into a handful of informative components.

e.g. PCA

Reinforcement Learning: Learning by Doing



Robotics example: a robot arm tries different grasp motions; successful grasps earn reward, and the policy improves through repeated trials — no one hand-writes the grasping rule.

Vocabulary You'll Use All Semester

Feature

A measured input variable (e.g., RMS vibration, temperature)

Label / Target

The correct output a supervised model is trained to predict

Training / Test Set

Data used to fit the model vs. held out to check it generalizes

Model

The learned function mapping features to predictions

Loss Function

A measure of how wrong the model's predictions are

X, y

Convention: X is the feature matrix, y is the target vector

How We'll Build Every Model This Semester

1
Preprocess
Clean, scale, and split the data

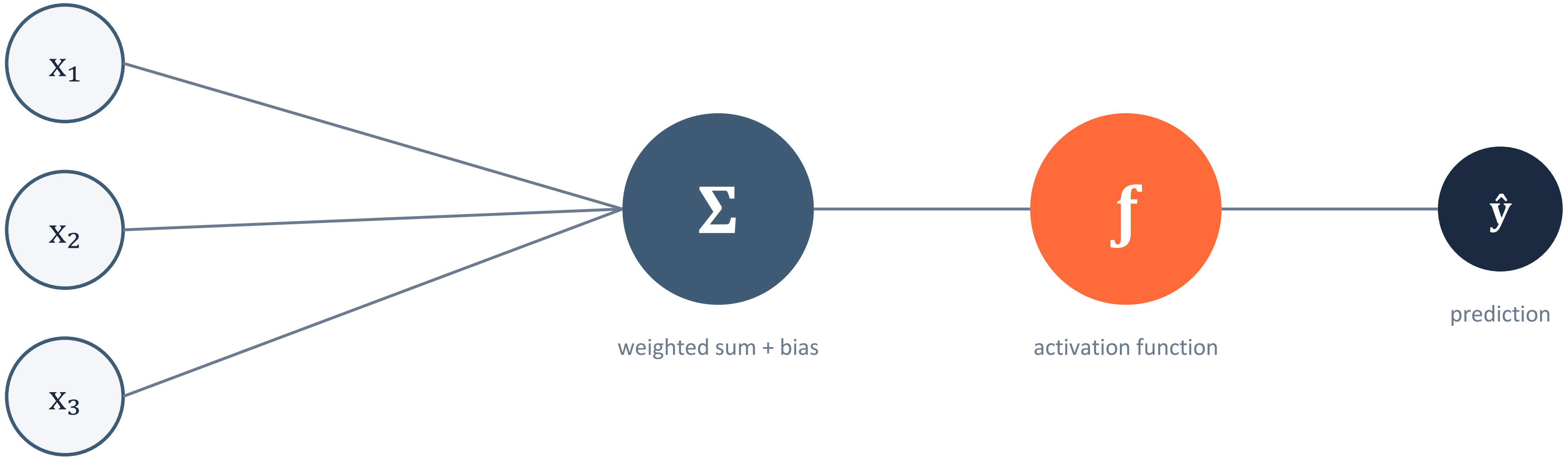
2
Train & Select
Fit candidate models, compare

3
Evaluate
Validate on held-out data

4
Deploy & Iterate
Use it, monitor it, improve it

The Building Block: The Artificial Neuron

Every network we build later this semester (Weeks 7–14) is, at its core, a stack of units like this.



1950s–60s: the Perceptron and Adaline. The full derivation is in Ch. 2 of the text — worth reading, but we'll move quickly through it here since Weeks 7–13 rebuild these ideas properly in PyTorch.

Your Toolbox This Semester

Python

The language for everything we do

NumPy

Arrays & numerical computing

Pandas

Loading & wrangling tabular data

scikit-learn

Weeks 1–6: classical ML

PyTorch

Weeks 7–14: deep learning

scikit-learn vs. PyTorch: When to Reach for Which

scikit-learn

- Tabular / structured sensor data
- Classical algorithms (trees, SVM, etc.)
- Fast to train, easy to interpret
- Your default for Weeks 1–6

PyTorch

- Images, sequences, large raw data
- Neural networks of any depth
- GPU-accelerated, highly flexible
- Your default for Weeks 7–14

Today's Lab: Setup & First Look at Sensor Data

1	**Verify your environment**	Confirm Python, NumPy, pandas, scikit-learn, and PyTorch all import correctly.
2	**Load a real sensor dataset**	Bring in vibration / IMU / force-torque data and inspect it with pandas.
3	**Visualize the signals**	Plot raw time series and compute simple summary features (RMS, mean, std).
4	**Train your first classifier**	Fit a logistic regression / perceptron to separate normal vs. faulty readings.

Key Takeaways

- ML complements classical control — use it where models are hard to derive by hand.
- Three paradigms: supervised, unsupervised, and reinforcement learning.
- scikit-learn carries Weeks 1–6; PyTorch takes over from Week 7 onward.

NEXT WEEK

A Tour of Classifiers & Data Preprocessing

Logistic regression, SVM, decision trees, k-NN — and cleaning the data you feed them.